ELECTRICAL AND COMPUTER ENGINEERING DEPARTMENT (ECE)

Faculty Members

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Vision

The Electrical and Computer Engineering Department (ECE) envisages being a leader in shaping intellectual and human capital in electrical, computer, communication, and biomedical engineering. The department seeks to produce application-oriented graduates in all fields of electrical, computer, communication, and biomedical engineering. Our graduates are trained to enrich Lebanon, the Middle East, and the world in general, technologically, economically, and socially.

Mission

The mission of the department is to be recognized as one of the best departments of electrical and computer engineering. Such excellence is not limited to education but should also encompass research, especially in translating research outcomes to the community at large.

Programs Offered

The ECE Department offers three programs – Electrical Engineering, Computer and Communications Engineering, and Biomedical Engineering. Each program grants three degrees: Bachelor of Science, Bachelor of Engineering, and Master of Science.

ELECTRICAL ENGINEERING PROGRAM

Mission

The Electrical Engineering program at RHU aims to help students explore their innate creativity and potential and endow them with the knowledge, skills and abilities to: pursue successful careers in electrical engineering and related fields; think critically in solving complex problems using modern tools and technologies; communicate and work effectively with diverse groups; learn more every day, and succeed in graduate studies in renowned institutions if they so choose.

Program Educational Objectives

The program's educational objectives are to enable graduates to:

1. Foster an environment that encourages excellence in endeavor.
2. Provide quality and professional education that prepares leaders.
3. Solve problems facing society and industry in Lebanon, the Middle East, and the world.
4. Engage in multidisciplinary research.

Student Outcomes

The Electrical Engineering program adopts the learning outcomes of ABET, so our graduates will have:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences.
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Accreditation

The Bachelor of Engineering program in Electrical Engineering is accredited by the Engineering Accreditation Commission of ABET (https://www.abet.org) under the commission's General Criteria and Program Criteria for Electrical and Similarly Named Engineering Programs.

Career Opportunities

An Electrical engineer could develop components for some of the most fun things in our lives (MP3 players, digital cameras, or roller coasters) as well as the most essential (medical tests or communications systems). This largest field of engineering encompasses the macro (such as large power grids that light up cities) as well as the micro (including devices smaller than a millimeter that trigger a car's airbags to inflate). An electrical engineer may work in robotics, computer networks, electric transportation systems, medical imaging, or renewable energy systems—at the very forefront of technological innovation—as well as in the low-power industry and power plants.

Program Overview

The Electrical (ELEC) Engineering Program at Rafik Hariri University puts students' best interests first. Every little experience the student gains represents a block in building a competent, confident, purposeful, problem-solving, competitive, responsible, and conscientious individual. This is accomplished through a curriculum and facilities that meet the highest standards, faculty members committed to students' academic and personal growth, and an environment that inspires learning and drives creativity.

The Bachelor of Engineering (BE) program in Electrical Engineering comprises 152 credit hours across eight regular semesters and three summer semesters. The Bachelor of Science (BS) program comprises 118 credit hours across six regular semesters and three summer semesters. In addition to completing the credit requirements, a BS or a BE degree is conferred upon a student if the earned Cumulative GPA, Major GPA, and the Summative Learning Project grade are all 70 or above.

The required credit hours span three categories: General Education requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours for the BS and the BE programs is shown in the following tables:

BS in Electrical Engineering (118 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education Requirement	0	0	20	7	27	23
College Requirement	3	0	18	3	24	20
Program Requirement	46	0	21	0	67	57
Credits	49	0	59	10	118	100

BE in Electrical Engineering (152 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	20	10	30	20
College Requirement	3	0	21	3	27	18
Program Requirement	56	15	24	0	95	62
Credits	59	15	65	13	152	100

I. General Education Requirements

The list of the GE required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

II. College Requirements

The list of the College-required courses and their descriptions is presented on the introductory pages of the College of Engineering section in this catalog.

III. Program Requirements

The program requirements for a Bachelor of Engineering degree in Electrical Engineering encompass 95 credits (67 credits for the BS degree), distributed as follows: 80 credits (67 credits for the BS degree) of mandatory courses and 15 credits of engineering technical electives for BE only.

A. Mandatory requirements

The mandatory courses for the major and non-major ELEC programs are listed in the table below.

Course #	Title	Credits	Prerequisites
CCEE 221	Logic Design	3	
CCEE 221L	Logic Design Lab	1	Pre-req.: CCEE 221

CCEE 331	Signals and Systems	3	MATH 216
CCEE 341	Communication Systems	3	CCEE 331
CCEE 426	Design of Embedded Systems	3	CCEE 221
CCEE 426L	Design of Embedded Systems Lab	1	Co-req.: CCEE 426
CCEE 460	Artificial Intelligence	3	CCEE 214
CCEE 460L	Artificial Intelligence Lab	1	Co-req.: CCEE 460
CCEE 534	Signal Processing	2	CCEE 331
ELEC 200	Introduction to Electrical and Computer Engineering	3	
ELEC 301	Sensors and Sensor Circuit Design	3	ELEC 210
ELEC 301L	Sensors and Sensor Circuit Design Lab	1	Co-req.: ELEC 301
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ELEC 323	Electronics Circuit Design	3	ELEC 301, ELEC 320
ELEC 330	Power Systems	3	ELEC 210
ELEC 331	Electric Machines	3	ELEC 210
ELEC 331L	Electric Machines Lab	1	Co-req.: ELEC 331
ELEC 340	Engineering Electromagnetics	3	PHYS 211
ELEC 400	BS Summative Learning Project*	3	ENGL 217
ELEC 405	Electrical System Design	3	ELEC 210
ELEC 406	Modeling and Simulation	3	ELEC 210
ELEC 431	Introduction to Renewable Energy Systems	3	ELEC 210
ELEC 432	Power Electronics	3	ELEC 320
ELEC 432L	Power Electronics Lab	1	Co-req.: ELEC 432
ELEC 451	Control Systems	3	CCEE 331
ELEC 499	Co-op Work Experience	3	ENGL 217
ELEC 533	Advanced Machines and Drives	3	ELEC 331, 432
ELEC 534	Advanced Power Systems	3	ELEC 330
ELEC 563	Smart Grid Technology	3	ELEC 432
ELEC 595A	BE Summative Learning Project 1	1	ENGL 217
ELEC 595B	BE Summative Learning Project 2	3	ELEC 595A
MECH 333	Thermal Fluid Sciences	3	MATH 216

* Required for BS-bound students. BE students take a technical elective instead.

B. Technical Electives

As part of the Bachelor of Engineering in ELEC Engineering program, the student is required to take 15 credit hours of engineering technical electives. Students must be of

senior standing to take engineering technical electives. Technical electives allow students to focus on a specific area to develop in-depth knowledge and understanding. The student can also mix and match elective courses from different areas to gain broader exposure to the various ELEC Engineering disciplines. The student should, in cooperation with their academic advisor, select a list of electives that best meet their needs and aspirations. The listed engineering technical elective courses and other courses from other engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in the following areas: (1) Power, (2) Renewable Energy, (3) Intelligent Systems.

It is highly recommended that students take elective courses from the list below after completing the mandatory requirements in their respective areas. Students may select technical electives from other programs, subject to approval of the academic advisor and the corresponding course instructor.

Course #	Title	Credits	Prerequisites
ELEC 433	High Voltage Engineering	3	ELEC 330
ELEC 434	Electrical Power Distribution	3	ELEC 330
ELEC 444	Electromagnetic Compatibility	3	ELEC 210
ELEC 498	Special Topics in Electrical Engineering	3	Instructor's consent
ELEC 503	Artificial Neural Systems	3	MATH 311
ELEC 504	Lasers and Laser Applications in Engineering	3	
ELEC 531	Electrical Power Systems for Transportation Applications	3	ELEC 432
ELEC 532	Application of Power Electronics in Power Systems	3	ELEC 330, ELEC 432
ELEC 535	Renewable Energy Systems for the Built Environment	3	
ELEC 536	Renewable Energy Systems	3	ELEC 431
ELEC 542	Advanced Power Electronics	3	ELEC 432
ELEC 551	Nonlinear Adaptive Control	3	ELEC 451
ELEC 560	Sustainable Energy	3	None
ELEC 561	Wind Energy Technology	3	ELEC 431 and ELEC 432
ELEC 562	Solar Energy Technology	3	ELEC 210
ELEC 597	Advanced Topics in Electrical Engineering	3	Instructor's Consent Senior Standing

C. Summative Learning Project SLP

Students must complete a 3-credit-hour course for BS (4 credits for BE; taken 1 credit in the first regular semester and 3 credits in the following regular semester) in which they work, preferably in groups, on a problem of concern to industry or the community at large, or to innovate a promising idea. The SLP could be an extension of the projects students had worked on in previous courses, bringing them to a more useful outcome.

D. Co-op Experience

Students must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on students' progress throughout the Co-op period by conducting field visits and ensuring that students' performance is aligned with their aspirations and employers' needs. Students are required to submit a formal report and a poster and to give a formal presentation about their Co-op experience. While ENGL 217 and Senior Standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Course Coding

The courses offered in the ELEC Engineering program are designated code numbers in the form of (ELEC abc), where:

a:	Designates course year/level
b:	Designates the focus area as follows: 0: General; 1: Electrical; 2: Electronics; 3: Power; 4: Magnetism; 5: Control
c:	Designates the course sequence in the area

For example, ELEC 320 is the *first Electronics course* offered by the *electrical engineering program* in the *second year*.

Study Plan

The following study plan summarizes the courses and credit distribution for the Bachelor of Science (BS) and Bachelor of Engineering (BE) in ELEC Engineering. The study plan serves as a roadmap to facilitate students' smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (14 Credits)			
ARAB 212	Arabic Language and Communication	2	
CCEE 214	Introduction to Programming	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
PHYS 211	Physics: Electricity and Magnetism & Lab	3	
Year 1, Spring Semester (15 Credits)			
CCEE 221	Logic Design	3	
ELEC 200	Introduction to Electrical & Computer Engineering	3	
ELEC 210	Electric Circuits	3	PHYS 191 or Equivalent
MATH 216	Calculus IV	3	MATH 215
PHYS 312	Modern Physics	3	

Year 1, Summer Semester (10 Credits)			
CCEE 221L	Logic Design Lab	1	CCEE 221
MATH 311	Linear Algebra and Applications	3	
	Humanities/ Fine Arts Elective I	3	
	Science Elective	3	
Year 2, Fall Semester (17 Credits)			
CCEE 331	Signals and Systems	3	MATH 216
ELEC 301	Sensors and Sensor Circuit Design	3	ELEC 210
ELEC 301L	Sensors and Sensor Circuit Design Lab	1	Co-req.: ELEC 301
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ENGL 217	Professional English	3	ENGL 210
MATH 351	Probability and Statistics	3	MATH 216
Year 2, Spring Semester (17 Credits)			
ELEC 323	Electronics Circuit Design	3	ELEC 301, ELEC 320
ELEC 330	Power Systems	3	ELEC 210
ELEC 331	Electric Machines	3	ELEC 210
ELEC 331L	Electric Machines Lab	1	Co-req.: ELEC 331
ELEC 340	Engineering Electromagnetics	3	PHYS 211
MATH 314	Ordinary Differential Equations	3	MATH 216
	Community and Sustainability Elective	1	
Year 2, Summer Semester (9 Credits)			
ENGR 300	Engineering Economics and Management	3	
MATH 421	Numerical Analysis	3	MATH 311
	Social Science Elective I	3	
Year 3, Fall Semester (16 Credits)			
CCEE 426	Design of Embedded Systems	3	CCEE 221
CCEE 426L	Design of Embedded Systems Lab	1	Co-req.: CCEE 426
ELEC 405	Electrical System Design	3	ELEC 210
ELEC 406	Modeling and Simulation	3	ELEC 210
ELEC 431	Introduction to Renewable Energy Systems	3	ELEC 210
ELEC 451	Control Systems	3	CCEE 331
Year 3, Spring Semester (17 Credits)			
CCEE 341	Communication Systems	3	CCEE 331
CEE 460	Artificial Intelligence	3	CCEE 214
CCEE 460L	Artificial Intelligence Lab	1	Co-req.: CCEE 460
ELEC 432	Power Electronics	3	ELEC 320
ELEC 432L	Power Electronics Lab	1	Co-req.: ELEC 432
ELEC	Engineering Technical Elective 1* or BS Project	3	
MECH 333	Thermal Fluid Sciences	3	MATH 216